The Renal Biopsy

25

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CHAPTER OUTLINE

KEY POINTS

- **Kidney biopsy plays an important role in the management of kidney disease.**
- **Percutaneous kidney biopsy is generally safe if care is taken to select and prepare the patients beforehand**
- **Full assessment of the biopsy requires examination by light microscopy, immunohistochemistry and (in most cases) electron microscopy**
- **The biopsy report should include a morphological description of the biopsy and an interpretation of the appearances in the light of the clinical presentation.**
- **New technologic developments in digital pathology, artificial intelligence, and (spatial) omics enable deep characterization of kidney biopsies toward precision kidney medicine.**

The renal biopsy has become a fundamental component in the management of renal disease. Its development and refinement since the late 1950s have been fundamental for the diagnosis and definition of clinical syndromes and the discovery of new pathologic entities.[1] Through the critical analysis of renal biopsies taken at different disease time points, key pathophysiologic features of kidney disease have been discovered, which have in turn helped to establish new paradigms in nephrology and have led to considerable alterations in patient management. Up to 74% of patients who are biopsied, management is estimated to be altered on the basis of biopsy results,[2] and in two thirds of patients, the biopsy reveals unsuspected diagnostic information. This is true for both native renal biopsies and renal transplant biopsies.[3] In addition, driven by development of novel technologies, much is being learned regarding disease pathogenesis through the study of renal biopsy material. Renal biopsy remains a “gold standard” for disease diagnosis and also allows the development of novel (biopsy) markers, which have revolutionized our concepts of pathologic mechanisms.

SAFETY AND COMPLICATIONS OF BIOPSIES

Although generally considered safe, there is a morbidity and small, but measurable, mortality associated with the procedure, so it is imperative to subject only those patients in whom there will be a potential benefit to these risks. Indications for renal biopsy may vary from one center to another, but accepted indications are listed in Box 25.1. The significant complications related to the procedure are hemorrhage, development of arteriovenous fistulas, and to a lesser extent sepsis. First, bleeding with macroscopic hematuria and the development of perinephric hematomas may be minor and self-resolving or major and require intervention in the form of blood transfusions, intravascular embolization, or rarely surgery (1.2%). Second, there is a risk of formation of arteriovenous fistulas (up to 14%),[4,5] which may be asymptomatic and spontaneously resolve or lead to a significant vascular steal syndrome, compromising the rest of the kidney through ischemia. Finally, there

Box 25.1 Indications for renal biopsy

- Proteinuria (significant proteinuria (>1 g/day or protein-to-creatinine ratio >100 mg/mmol)
- Acute nephritic syndrome (microscopic hematuria with any degree of proteinuria)
- Unexplained reduction in kidney function (native or transplant kidney)
- Renal manifestations of systemic disease

is the risk of sepsis following the procedure, through the introduction of a septic focus or its dissemination. Overall, the risks of complication vary from center to center and between practitioners but can be estimated between 3.5% and 13%, with the majority being minor complications (approximately 3%–9%),[6–8] although this appears to have decreased in recent years.[9,10] Mortality from the procedure is generally a result of undiagnosed bleeding with significant hematoma formation and was reported in up to 0.2% of cases from some of the larger biopsy series,[9] although other studies suggest that it represents an extremely rare adverse event.[11] Some degree of bleeding is common, as approximately half of patients have a small drop in hemoglobin post biopsy, and one third will develop some hematoma, but only in a minority (≤7%) will bleeding be significant and require intervention.[12] Complications appear to be more common in native than transplant kidneys and in patients with more advanced renal impairment, with prolonged bleeding times, or with lower hemoglobin.[7,8] In a recent retrospective study (1995–2015), complications after 1705 percutaneous native and transplant kidney biopsy were compared revealing a greater drop in hemoglobin (0.97 vs. 0.73 d/dL), a higher complication rate (6.5% vs. 3.9%), and higher transfusion rate (5.2% vs. 3.3%) after native kidney biopsies. One prospective study identified the only risk factors for bleeding complications as being female, younger (35 ± 14.5 years vs. 40.3 ± 15.4 years), and having a prolonged partial thromboplastin time.[13] This finding was recently confirmed in a retrospective analysis of the Boston Kidney Biopsy Cohort, which revealed that the post-biopsy transfusion risk was associated with baseline hemoglobin level and female sex; estimated GFR and baseline hemoglobin were independently associated with a larger drop in hemoglobin. Other retrospective univariate analyses have reported that blood pressure of 160/100 mm Hg or higher or a serum creatinine of greater than 2 mg/dL more than doubled the risk of bleeding.[14] Overall, however, no effective means has been established to identify those individuals at risk of developing "clinically" significant complications, while risk calculators (e.g., http://perioperativerisk.com/kbrc) have been suggested to assess the personalized risk, their use has not been evaluated in RCTs. In one small series, post-biopsy ultrasound within an hour had a 95% negative predictive value for predicting clinically significant hemorrhagic complications,[12] meaning that the absence of a hematoma on the post-biopsy scan was suggestive of an uncomplicated clinical course.

Debate continues regarding the routine use of DDAVP to counteract uremic bleeding tendencies. In part, this is because its use was previously reserved for only those patients with prolonged bleeding times, and numerous studies have since demonstrated that complication rates are no different if bleeding time estimation is omitted from the preoperative assessment,[15,16] as it does not predict clinical complications.[13] However, more recent data from a randomized double-blinded trial suggested a significant benefit in preventing bleeding complications with few adverse events.[17] A total of 162 low-risk adult patients undergoing biopsy were enrolled and randomized to subcutaneous DDAVP (0.3 µg/kg) or placebo. The patients were normotensive and had preserved renal function with serum creatinine of <1.5 mg/dL (estimated glomerular filtration rate >60 mL/min). Those who received DDAVP evidenced a significant reduction in incidence of post-biopsy bleeds (30.5% vs. 13.7%; relative risk 0.45), a significant reduction in hematoma size in those who did bleed, and a reduction in duration of hospital stay. However, hemoglobin drop after biopsy was minimal and there were no major complications, leading some to question the benefit of reduction in clinically unimportant hematomas, which can be frequently found following biopsy if looked for. No thrombotic, hyponatremic, or cardiovascular events were recorded. Whether these data, in patients with preserved renal function, could be translated to those higher-risk patients with greater renal impairment is unclear and is a question worthy of a further randomized trial.

Many centers stop antiplatelet therapy prebiopsy for elective procedures, but recent data suggest that bleeding rates may be no different in those taking aspirin or stopping a week beforehand, and continuing treatment may avoid the increased risk of cardiovascular events following aspirin withdrawal. In a center that does not routinely stop aspirin (but does stop clopidogrel), a retrospective analysis of 2563 biopsies revealed a major bleeding complication in only 2.2%, and in those in whom a complete drug record was available, no significant difference was noted on or off aspirin.[10] Limited data are available on biopsies performed on patients taking clopidogrel.[18]

Along with developing procedure-related complications, there is the chance that an inadequate core of tissue is obtained for diagnosis, containing too few glomeruli or insufficient cortical material, and this is reported in between 1% and 5% of cases. The size requirements for accurate diagnosis are discussed later.

Some absolute contraindications preclude percutaneous biopsy, whereas relative contraindications (Table 25.1) may be tolerated depending on the importance of the biopsy, operator's experience, and supportive facilities available. Ideally, all efforts should be made to address relative contraindications; however, in the context of acute kidney injury and rapidly progressive glomerulonephritis or suspected transplant rejection, this may not always be possible. With modern techniques, evidence is emerging that previously perceived high-risk factors such as obesity, plasma cell dyscrasias such as myeloma, or amyloidosis are not actually associated with higher rates of bleeding complications.[19] The critical preoperative steps are to ensure that blood pressure is controlled, the patient does not have a bleeding diathesis or a urinary tract infection, and the kidneys are suitably imaged, with no evidence of obstruction, widespread cystic disease, or malignancy (although percutaneous biopsy is increasingly used to diagnose the nature of renal masses). As a result, preprocedural assessment should allow those patients unsuitable for percutaneous biopsy to be referred for an alternative approach (Table 25.2). In these patients, there are other means of obtaining renal tissue, which include open biopsies,[20] laparoscopic biopsies, or transjugular biopsies.[21] Each is associated with specific complications and has particular merits depending on the clinical scenario (see Table 25.2). These alternative approaches are generally only required for a minority of patients requiring biopsies.

The safe duration for observation following renal biopsy has been investigated in numerous studies, which suggest that early discharge (after only 4-hour observation) will result in a number of missed complications, with many more occurring between 8 and 24 hours post procedure. Even after 8 hours, 23% to 33% of complications will be missed. However, an overnight stay will allow an extra 20% of complications to be identified before discharge with between 85% and 95% of complications being identified at 12 hours and 89% and 98% following 24-hour observation.[7,22] Some units practice a policy of day biopsies with a minimum 6-hour bed rest period, which is extended only if there is evidence of bleeding, and this appears to be associated with no increased complication rates.[23] Vigilant observation of blood pressure, pulse rate, and evidence of hematuria is required in all cases.

Table 25.1 Contraindications to renal biopsy

Absolute contraindications	Relative contraindications
Uncontrolled hypertension	Single kidney
Bleeding diathesis	Antiplatelet/clotting agents[a]
Widespread cystic disease	Anatomic abnormalities
Hydronephrosis	Small kidneys
Uncooperative patient	Active urinary/skin sepsis
	Obesity[a]

[a]Aspirin and body mass index >40 kg/m^2 were not found in recent limited cohort data to be a significant risk.

BIOPSY HANDLING

Detailed descriptions of methods of handling biopsies can be found in many publications. A full assessment of the renal biopsy requires examination by light microscopy, immunohistochemistry, and electron microscopy (EM, also referred to as "triple diagnostics"), with the use of other tests in some circumstances. Light microscopy and immunohistochemistry are required for all cases, whereas EM is required for some diagnosis but could be omitted in other cases (see later). Therefore it is necessary to divide the biopsy tissue for each of these methods of examination. During this process, it is extremely important that the tissue is not damaged by handling, or by drying, and that the tissue is fixed in an appropriate fixative as quickly as possible, ideally within minutes. This is best achieved by dividing the biopsy at the bedside. Examination of the biopsy with a dissecting microscope allows the cortex, containing glomeruli, to be distinguished from medulla and nonrenal tissue (such as fat) (Fig. 25.1). This facilitates assessment of the adequacy of the cores and division of the biopsy so that glomeruli are present in the samples for each modality of examination. If a dissecting microscope is not available, then a standard light microscope can be used with the biopsy placed in a drop of normal saline on a microscope slide. If it is not possible to examine the biopsy in this way, then a standard approach to

Table 25.2 Alternative methods for obtaining renal tissue and their risks and benefits compared with a percutaneous approach

Method	Advantage	Disadvantage
Transjugular approach	Can be of use in those with a bleeding diathesis, ventilated patients, or if combined liver and renal biopsy is required	Risk of renal capsule perforation Inadequate material in up to 24%
Open approach	High yield of adequate tissue Hemostasis is more secure	Requires general/spinal anesthesia; longer recovery period
Laparoscopic approach	High yield of adequate tissue Hemostasis is more secure	Requires general/spinal anesthesia; longer recovery period

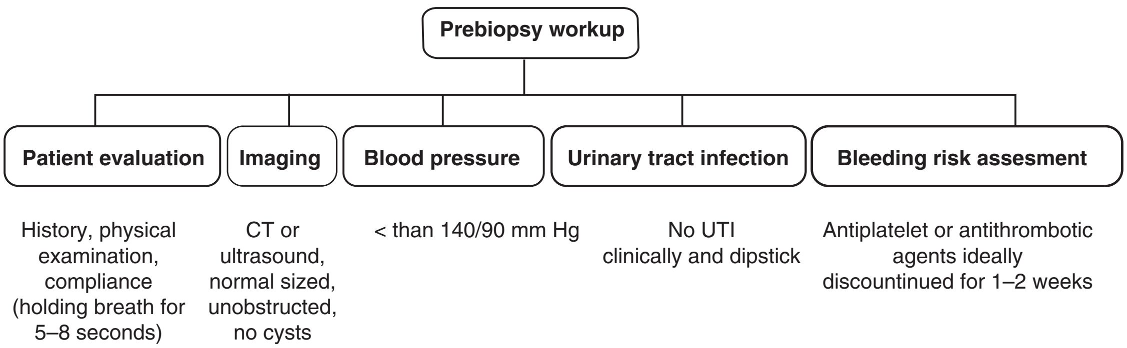

Fig. 25.1 Prebiopsy flowchart. *CT,* Computed tomography; *UTI,* urinary tract infection.

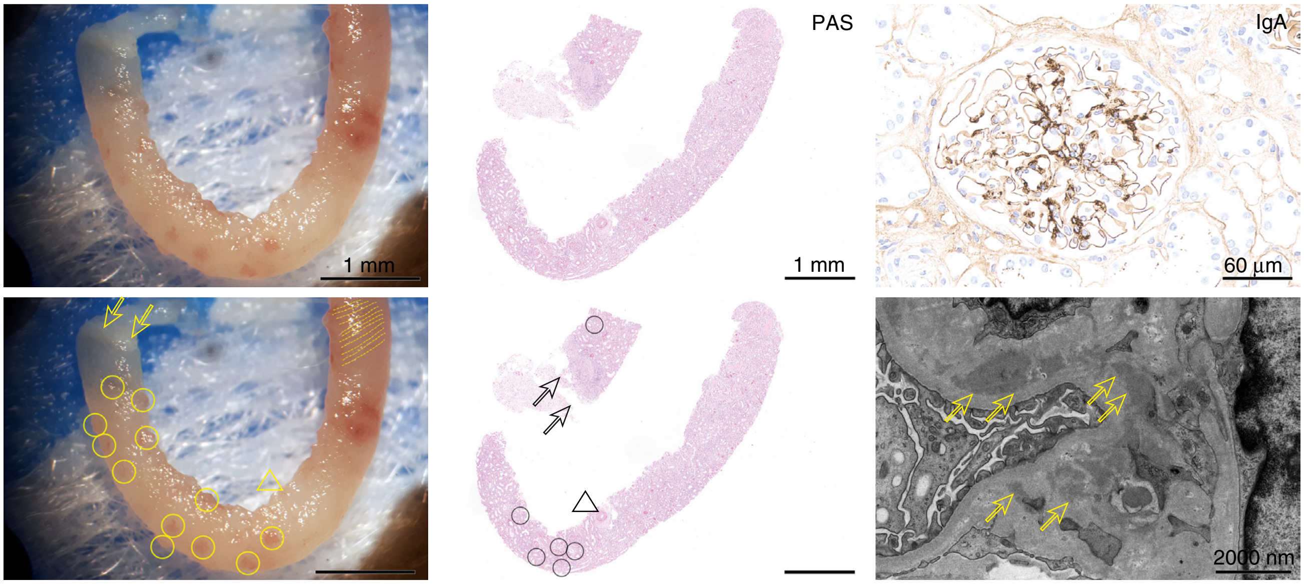

Fig. 25.2 Macroscopic appearance of kidney biopsy under a dissection microscope and its histopathologic workup on a case of IgA nephropathy. Under the stereomicroscope, the thick fibrous renal capsule (*arrows,* left image) is a good indicator of the cortical portion of the biopsy. It includes the glomeruli, which are visible as red spheres (*circles,* lower left image) in the fresh biopsy. The opposite part of the biopsy may contain intrarenal arteries *(arrowhead)* and medullary tissue indicated by *red lines* as a morphologic correlate of the vasa recta (*dashed lines,* bottom left image). For electron microscopy, a 1 × 1 × 1 mm piece of cortical tissue including glomeruli is removed and processed separately. The remaining tissue is dehydrated overnight. The dehydrated tissue is embedded in paraffin and cut into consecutive 1-μm thick cuts, mounted on a microscope slide. Slides are stained with various histochemical stainings including periodic acid–Schiff stain (PAS, middle images), highlighting glomerular basement membranes (*circles,* lower middle image), tubular basement membranes, and arterial walls *(arrowhead).* The subcapsular area (*arrows,* lower middle image) is separated from the biopsy core due to sampling for electron microscopy. Additionally, immunohistochemistry is applied to characterize immunoglobulin (e.g., mesangial IgA as shown by the dark brown color and upper right image) and complement deposition within glomeruli. Electron microscopy is used to detect electron-dense deposits (*arrows,* lower right image).

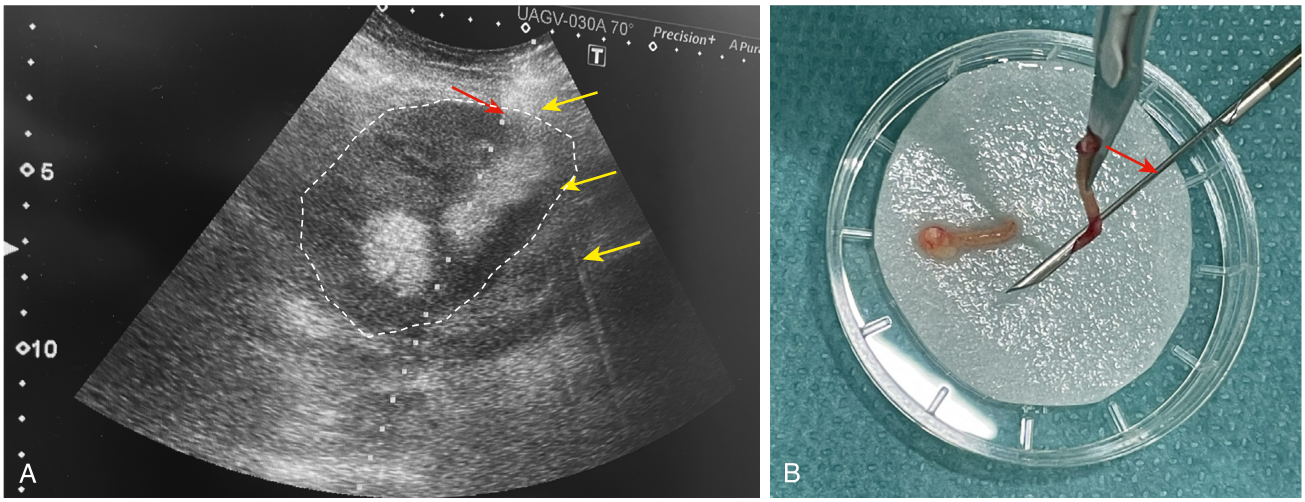

Fig. 25.3 Ultrasonography image during biopsy procedure and macroscopic appearance of biopsy. (A) Ultrasonography of a transplant kidney biopsy. The "flash" artifact from the biopsy needle is indicated by *yellow arrows,* while the tip of the biopsy needle is marked in *red.* (B) Macroscopic view of a kidney biopsy specimen retrieved from the biopsy needle (marked in *red*).

obtain material for EM is to take small fragments (≈1 mm in length) from each end of each core. In that way, if there is cortex in the core, glomeruli should be sampled. The remainder of the cores can be either all fixed or divided for light microscopy and immunofluorescence (IF). The part of the biopsy for light microscopy is then placed in appropriate fixative, most often formalin, and the part for IF is either snap frozen or transported to the laboratory in a suitable transport medium such as that described by Michel and colleagues.[24] Tissue placed in this medium can remain for several days at room temperature without loss of antigens. During division of the biopsy, it is important not to introduce artifacts due to crushing or stretching. Forceps should not be used to pick up the specimen; small polyester cleaning swabs are useful, but if not available, gentle use of a needle or a small wooden stick such as a toothpick is also possible. The biopsy should be cut using a fresh scalpel. If the biopsy must be taken to the histology laboratory for division, this should be done as quickly as possible with the biopsy wrapped in saline-moistened gauze or tissue culture medium. Artifacts may be produced if the biopsy is placed on dry gauze or gauze moistened with water or if it is placed in ice-cold saline. Some nephropathology laboratories use immunohistochemistry on formalin-fixed tissues, not requiring sample division, with a single biopsy core being sufficient for the diagnosis in most cases. It is also possible to take samples for EM from formalin-fixed biopsies such that the whole biopsy core is sent in one vial with formalin and all further workup is done by the nephropathology laboratory. This makes the process easier for the biopsy-performing centers, particularly those without in-house pathology.

If the amount of material obtained at biopsy is limited, then it may be necessary to adapt the way in which it is divided. The decision as to how this is done must depend on the clinical question. This procedure is preferentially performed by the nephropathology laboratory due to experience in tissue handling. In most cases, it is possible to omit frozen material for IF and instead perform immunohistochemistry on paraffin sections. However, if there is a suspicion of crescentic glomerulonephritis due to antiglomerular basement membrane (anti-GBM) disease, IF is more reliable in detecting the linear capillary wall staining. It may be possible to omit EM or perform EM on material reprocessed from the paraffin block. However, accurate measurements of glomerular capillary membrane thickness are not reliable after this procedure.[25] Cost and method availability should also be considered in the biopsy workup (e.g., EM is not available in all pathology laboratories and is costly but not required for diagnosis in most transplant biopsies). Therefore the need for EM can be considered on a case-to-case basis.

Box 25.2 Features to be assessed by light microscopy in glomeruli

- Globally sclerotic glomeruli
- Size
- Cellularity: increased cells (hypercellularity) in mesangium (mesangial hypercellularity; NOTE: normal mesangial areas contain two to three cells), in capillary lumens (endocapillary hypercellularity), or in Bowman space (extracapillary hypercellularity or crescents)
- Capillary wall thickness and form (e.g., wrinkled, with double contours, spikes, spicules, craters, and holes; NOTE: best seen in periodic acid–Schiff or silver stain)
- Mesangial expansion (mesangial sclerosis) with/without nodules
- Deposition of abnormal material (e.g., amyloid)
- Segmental sclerosis
- Presence of thrombi, pseudothrombi, necrosis
- The distribution of the previously mentioned lesions (i.e., focal vs. diffuse and segmental vs. global)

Box 25.3 Features to be assessed by light microscopy in tubules

- Percentage atrophy
- Signs of acute damage (e.g., dilatation, loss of brush border, epithelial flattening, mitoses, and necrosis)
- Tubulitis
- Casts: Granular casts suggest acute tubular injury; eosinophilic fractured casts suggest myeloma; neutrophil casts suggest acute pyelonephritis, pigmented cast (e.g., bile acids and myoglobin)
- Crystals (e.g., oxalate and calcium phosphate)
- Viral inclusions (e.g., BK virus)
- Distribution of the abovementioned lesions (i.e., focal vs. diffuse)

LIGHT MICROSCOPY

Most renal pathologists employ several stains for light microscopy. Commonly used stains are periodic acid–Schiff (PAS) reaction, hematoxylin-eosin (H&E), silver methenamine, and trichrome. It is not necessary to perform all these stains. PAS and H&E are used by most laboratories. The PAS stain, also referred to as "H&E for the kidney," is the most widely used and useful staining in nephropathology, permitting a good overview and visualization of details on kidney architecture.

Kidney biopsies must be examined systematically, first at low power to identify parts of the kidney (or other structures in some cases), whether there is cortex and/or medulla, the amount of chronic nephron damage with tubular atrophy and interstitial fibrosis, and the presence of interstitial inflammatory infiltrates. This will also permit assessment of interstitial expansion, most commonly due to either edema or fibrosis, but occasionally due to infiltration by, for example, amyloid. Although seldom seen in nephrology-indicated kidney biopsies, the presence of neoplastic nonmalignant or malignant lesions should also be assessed. Examination should then proceed by studying all other compartments in more detail (i.e., glomeruli, tubules, interstitium, and vessels including arteries, arterioles, and veins). Features that should be looked for in glomeruli and tubules are detailed in Boxes 25.2 and 25.3. Arterioles should be examined for the presence of hyalinosis, thrombosis, and necrosis. Arteries should be assessed for intimal thickening and whether it is accompanied by reduplication of the internal elastic lamina, thrombosis, necrosis, inflammation, and cholesterol emboli.

TERMINOLOGY IN DESCRIPTION OF KIDNEY DISEASES

The involvement of glomeruli by a pathologic process can be defined by the percentage of glomeruli involved by a lesion and by whether the lesion involves all or only part of any individual glomerulus. A lesion that involves all or nearly all glomeruli is described as "diffuse," whereas one that involves some but not all glomeruli is described as "focal." In the definitions given in the World Health Organization (WHO) atlas of glomerular diseases, it was suggested that the cutoff for focal versus diffuse should be 80% of glomerular involvement. However, in recent classifications of lupus glomerulonephritis[26] and immunoglobulin A (IgA) nephropathy,[27] the cutoff is defined as 50%. If a lesion involves only part of a glomerulus (i.e., with some capillary lumens remaining uninvolved), it is called "segmental," whereas if it involves the whole glomerulus, it is called "global." In the classifications of lupus glomerulonephritis and IgA nephropathy, the cutoff is set at 50% glomerular tuft involvement, except for segmental sclerosis in IgA nephropathy, in which any area of sclerosis that leaves some of the glomerulus unaffected is defined as segmental. Specific terminology is also used for tubulointerstitial and vascular lesions. A consensus on terminology of kidney lesions and their definitions has been recommended by the Renal Pathology Society.[28]

IMMUNOHISTOCHEMISTRY

Several diagnoses cannot be made without immunohistochemistry including IgA nephropathy, C3 glomerulopathy (GN), anti-GBM disease, and light-chain deposition disease. In native kidneys a minimum panel includes antibodies for IgA, IgG, IgM, C3c, and kappa and lambda light chains. Light-chain immunohistochemistry is important if diagnoses such as light-chain deposition disease, monoclonal immunoglobulin deposition disease, or proliferative glomerulonephritis with monoclonal immunoglobulin deposits are not to be missed (but could be omitted in children and young adults in which such diseases are extremely rare). Many pathologists would also add antibodies for C1q, C4c, and fibrinogen to this routine panel. In transplant kidney biopsies a standard panel would normally include C4d and SV40. C4d is invaluable in assessing the activation of the classical pathway of complement by antibodies and is required in the diagnosis of antibody-mediated rejection. SV40 antibody detects the polyoma SV 40 T-antigen and is helpful in diagnosis of BK nephropathy (named after a patient whose initials were B.K.). Detection of several other antigens may be useful in particular circumstances:

1. Microorganisms, in addition to BK virus, cytomegalovirus, and Epstein-Barr virus.
2. Amyloid proteins. Antibodies are available to AA and AL amyloid and many rarer inherited forms of amyloid.
3. α chains of type 4 collagen. In suspected hereditary nephropathy of Alport type, it may be helpful to stain for the α3 and α5 chains of type IV collagen.
4. IgG subclasses in cases of suspected monoclonal immunoglobulin deposition.
5. Myoglobin in suspected myoglobinuria.
6. Lymphocyte surface antigens, particularly in cases of suspected lymphoid neoplasia.
7. Type III collagen in collagenofibrotic GN.
8. Fibronectin in fibronectin GN.
9. Disease-relevant antigens (e.g., PLA2R1 [and several others] in membranous nephropathy).
10. DNAJB9 in fibrillary GN.

In reporting immunohistochemical staining for immunoglobulins and complement, it is important to describe the site of staining in the glomerulus (e.g., mesangial or capillary wall) and its nature (whether linear, finely, or coarsely granular). Intensity should also be described, but this is only reliably possible when using immunofluorescence. For estimation of intensity, most pathologists rely on semiquantitative subjective scale from 0 to 3+, but formal quantitation by image analysis may be useful for research. In addition to the glomerulus, staining should also be assessed in the tubules, especially the tubular basement membrane, interstitium, and vessels.

ELECTRON MICROSCOPY

EM is invaluable for assessing structural changes in the glomerulus and for identifying immune complexes, which are seen as areas of electron density and sometimes show an organized appearance. As with light microscopy, examination of the biopsy by EM should be systematic with assessment of the glomerular capillary basement membrane and its thickness; the endothelium and whether there is thickening or loss of fenestrations; the capillary lumen and particularly whether there is narrowing by cells or other material; the podocytes, looking particularly at the preservation of the foot processes and whether the cell bodies show any vacuolation or microvillous change. The presence of any electron-dense deposits—most commonly due to immune complex deposition—should be noted together with their distribution: mesangial, subendothelial, or subepithelial. EM may also demonstrate several other structures such as fibrils in amyloidosis or fibrillary glomerulonephritis, tubules in immunotactoid GN, or the characteristic inclusion bodies of various storage diseases.

Although EM is most useful in the assessment of glomerular morphology, it may also be helpful in demonstrating ultrastructural changes in other parts of the kidney. For example, it may help in demonstrating tubular basement membrane immune complexes, elucidating the nature of tubular epithelial cell inclusions, and examining the morphology of mitochondria in tubular epithelial cells, which may show abnormalities in inherited conditions or as a result of drugs.

Most studies suggest that EM provides useful information in about half of all native kidney biopsies and is essential for diagnosis in about 20%.[29] Because it is impossible to know which these are at the time of biopsy, it is prudent to always have material available for EM even if, in some cases, it is not processed further after light microscopy and immunohistochemistry. Box 25.4 lists some conditions for which EM is essential for the diagnosis and others where it is helpful. Also listed are some conditions in which the diagnosis may be reached without EM, but even in these cases, it is important to remember that EM may allow a more detailed description of the morphology of these conditions or also reveal a totally unrelated pathology.

Morphometric analysis of EM is mainly of importance for research. However, it is important to be able to measure the thickness of the GBM to quantify the thinning that may be seen in Alport syndrome or the thickening commonly seen in diabetic nephropathy. Accurate, unbiased measurement of the GBM thickness requires complex morphometric techniques to avoid the bias introduced by tangential sectioning of capillary loops. However, in practice, it is

Box 25.4 Examples of the use of electron microscopy (EM) in diagnosis of kidney biopsies

EM essential for diagnosis

Immunotactoid glomerulonephritis (GN)
Alport syndrome (including thin basement membrane lesion)
Fabry disease
Lecithin-cholesterol acetyltransferase deficiency
Nail patella syndrome
Fibrillary glomerulopathy (if DNAJB9 immunohistochemistry is not available)

EM helpful

Dense deposit disease
Minimal change disease
Early diabetic GN
Early membranous GN (particularly if only paraffin sections are available for immunohistochemistry)
Membranoproliferative GN
Postinfectious GN
Human immunodeficiency virus nephropathy
Lipoprotein GN
Collagenofibrotic GN

Diagnosis may be made without EM

IgA nephropathy
Acute tubulointerstitial nephritis
Myeloma cast nephropathy
Pauci-immune crescentic GN
Amyloid (although amyloid fibrils may be detected by EM when it has been missed on light microscopy)

satisfactory to use direct measurement of GBM thickness (distance from endothelial to podocyte plasma membrane) and determination of the arithmetic mean of such measurements. Reproducible results can be reached by 16 measurements from each of 2 glomeruli using this direct method.[30] Ideally, each laboratory should define a normal range using this method.

OTHER STUDIES ON THE RENAL BIOPSY

Other methods can be used for studying kidney biopsies. In particular, molecular pathology methods can be useful (i.e., analysis of DNA and RNA, which has become standard practice in tumor pathology). In cases of suspected kidney infection, part of the biopsy may be sent for culture. If this is not available, DNA or RNA might be isolated (even from a single histology slide and after microdissection) for polymerase chain reaction to detect DNA or RNA of infective organisms. This can be particularly helpful if no reliable immunohistochemistry is available or immunohistochemistry results are ambiguous. In biopsies with lymphoid infiltrates, immunoglobulin gene rearrangement analysis is used to confirm clonality and thus the malignant nature of an infiltrate.

Extracting messenger RNA (mRNA) from biopsies enables the study of differences in gene expression in different pathologic conditions.[31] These techniques have been applied to whole biopsies or parts of the biopsy (e.g., glomeruli isolated either by simple dissection under a dissecting microscope or by laser capture microdissection). Examination of mRNA transcripts shows considerable promise for assisting in the diagnosis of kidney transplant rejection and is now included in the Banff classification.[32]

The analysis of the range of proteins—the proteome—in the biopsy (also denoted as tissue proteomics) is evolving as another promising approach.[33] Using mass spectrometry to analyze the proteome may allow identification of the fibril proteins of amyloid or fibrillary glomerulonephritis and is already used in biopsy diagnostics in a few centers.[34,35] Furthermore, the chemical composition of unclear material in the biopsy (e.g., crystalline material) may be determined by energy-dispersive x-ray spectroscopy, mostly available with EM instruments.

SIZE OF THE BIOPSY

The renal biopsy is only a small sample of the renal parenchyma and this always needs to be kept in mind when making inferences about the state of the whole kidney from changes seen in the biopsy. Some diseases may only affect the kidney focally and therefore may be missed on biopsy (e.g., reflux nephropathy or arterial cholesterol emboli). Others may be segmental at the level of the glomerulus, such as focal and segmental glomerulosclerosis or pauci-immune necrotizing glomerulonephritis, and therefore the chance of detecting them will depend on how many glomeruli are present in the biopsy and how many sections are examined. Sampling is also a problem when we make inferences from the extent of disease we see in the biopsy to the extent that affects the kidney. For example, if 20% of the glomeruli in a biopsy have crescents, we tend to assume that this is the percentage of glomeruli in the kidney that have crescents. However, because of the small size of most biopsies, the confidence limits we can place on the true involvement of glomeruli are usually wide. An elegant mathematical description of the problems of glomerular sampling showed, for example, that to confidently exclude a segmental glomerular disease that is affecting about 5% of the glomeruli, a biopsy with 20 glomeruli is needed.[36] The situation is worse if we consider the problem of comparing the amount of glomerular involvement in two different biopsies, a question that often arises, for example, in patients with lupus glomerulonephritis who have repeat biopsies. In that case, to confidently detect a 10% difference in glomerular involvement between two biopsies would require more than 100 glomeruli in each biopsy. To detect differences of 25% to 40% glomerular involvement, the minimum biopsy size is 20 to 25 glomeruli.

For some diseases, classification schemes have defined minimum sizes for biopsy adequacy. Thus for lupus glomerulonephritis, it is suggested that a biopsy should contain a minimum of 10 glomeruli.[26] In transplant biopsies, the Banff group[37] has suggested that requirements for biopsy adequacy are 10 or more glomeruli with at least 2 arteries. It has been shown that examining two rather than one core of tissue increases the sensitivity for the diagnosis of acute rejection from 91% to 99%.[38] In acute cellular rejection, examining slides taken at only one level, rather than at three, misses 33% of cases with intimal arteritis.[39]

Therefore the majority of nephropathology laboratories use multiple sections per case.

THE BIOPSY REPORT

The biopsy report should include a morphologic description of the biopsy and an interpretation of the appearances in light of the clinical presentation. The changes seen in light microscopy, immunohistochemistry, and EM must be integrated, and this is best done if a single person examines the biopsy by each modality. A Committee of the Renal Pathology Society has published recommendations for the essential elements that should be present in a renal biopsy report.[40]

The description of light microscopy should include the number of glomeruli present and number that shows global or segmental sclerosis. It is essential to provide a quantitative estimate of the amount of irreversible nephron damage in the biopsy and, where appropriate, the severity of any active inflammatory process. The best way to estimate the irreversible damage is by specifying the number of globally sclerosed glomeruli and the amount of tubular atrophy and interstitial fibrosis. The estimate of activity will depend on the particular disease process but should include an indication of the proportion of glomeruli involved by crescents, necrosis, and endocapillary hypercellularity. For some diagnoses, there are established classification schemes that should be applied to the biopsy (e.g., International Society of Nephrology/Renal Pathology Society classification of lupus nephritis,[26] Oxford Classification of IgA nephropathy,[27,41] Banff classification of allograft pathology). International consensus groups have published recommendations for Pathologic Classification, Diagnosis, and Reporting of GN[42] and for the standardized grading of chronic changes in native kidney biopsy specimens.[43]

The interpretation of renal biopsies requires the pathologist to integrate biopsy findings with detailed clinical information and therefore requires a thorough understanding of renal disease and therapeutic implications of the biopsy diagnosis. Close communication between the clinician and pathologist is essential, and it is generally helpful for the biopsy to be viewed and discussed at a clinicopathologic conference. This is now facilitated by virtual meetings and digital pathology enabling telepathology.

EMERGING TECHNOLOGIES AND NEXT-GENERATION BIOPSY ASSESSMENT

A breakthrough in pathology diagnostics is being enabled by digital pathology. This mainly refers to the digitization of the glass slides using high-throughput whole-slide scanners, generating the so-called whole-slide images (WSIs). Currently, WSIs are among the largest medical datasets generated, with gigapixel resolution and sizes of up to several GB per single WSI. This workflow enables the pathologists to view and analyze the histopathology on computer screens using specialized software (like radiology). Digital pathology has several key advantages: telepathology: the digital slides and whole diagnostic can be performed remotely, facilitating accessibility; flexible working conditions like a home office; consultations and second opinions; and international collaborations. Because digital slides can be more easily archived, retrieved, and shared and they do not fade/degrade with time, they allow more efficient workflows and time savings. On the other hand, the necessary digital infrastructure requires significant investment and maintenance. Easy access to digital slides facilitates education and training for medical students, residents, and pathologists. Digital pathology also improves quality control and accuracy, potentially reducing errors. Importantly, digital pathology opens unprecedented possibilities for advanced image analysis and use of artificial intelligence, particularly deep learning (i.e., computational pathology).

Most developments in computational pathology are using end-to-end deep learning models to predict clinically actionable readouts from histopathology, mainly in cancer. These include disease classification, progression, molecular alterations, or treatment response. Such end-to-end approaches are relatively easy to train (if sufficient annotated data are available) and use, given that they provide categorical outputs that are relatively easy to interpret (e.g., progressor vs. nonprogressor). However, such models have limited explainability, also referred to as a "black box" (i.e., it remains unclear how the deep learning model performs its decisions).

Digital histologic images contain a wide range of features that can be analyzed. These include sizes (e.g., area and length of axes); shapes (e.g., circularity, eccentricity, and elongation); colors (e.g., hue and intensity); or textures (e.g., homogenous and heterogeneous). Other features can describe the relationship of histologic structures or cells to each other (e.g., distance between similar structures, such as glomeruli or tubules) and their orientation at the tissue level (e.g., density or orientation). Recent developments now enable an unbiased, comprehensive, and automated extraction of features from histologic structures or cells on a large scale. This approach has been termed "pathomics" and the technology is called "next-generation morphometry" because it resembles and seamlessly extends other omics technologies, providing the missing analyses of structural changes in tissues.[44,45] Pathomics uses histological slides already available from clinical routine or preclinical research and can be run on standard computational infrastructure. This enables relatively easy implementation in research and clinics compared with many other molecular omics approaches, which often require specific, costly infrastructure and sample preparation. Pathomics features have been shown to predict disease progression in patients with IgAN, derive new findings in patients with minimal change diseases, and enable computational reconstruction of morphologic disease progression in chronic kidney disease.[45] Pathomics is also valuable in preclinical research, enabling large-scale quantification. For example, this facilitated the analysis of disease regression in experimental models and served as a surrogate endpoint for a preclinical randomized controlled trial.[46,47] Breakthroughs in molecular omics technologies have led to a new era of molecular phenotyping, allowing for deep and comprehensive assessments of human tissues including kidney biopsies. These promising techniques include single-cell RNA sequencing and spatial multiplex

RNA and protein imaging, the latter allowing deep molecular analysis in conjunction with the assessment of morphologic tissue organization. These methods are increasingly becoming standard analytic approaches in kidney research, revealing important insights into disease pathophysiology.[48–50] The potential integration into routine clinical biopsy diagnostics is being assessed and will likely become available when the technologies become more broadly available and affordable. Use of deep learning can facilitate mapping of highly multiplexed protein or RNA tissue measurements with routine H&E or PAS images, unlocking deeper, previously inaccessible insights into tissue pathology.

Other approaches enabling novel and deeper insights into kidney pathophysiology include nondestructive and 3-dimensional imaging techniques. In the long term, nondestructive imaging (e.g., using hierarchical phase-contrast tomography [synchrotron-based imaging])[51] could potentially replace the need for conventional histopathology of biopsies, sparing the precious tissue for detailed molecular analysis.

All these methods enable a deep characterization of kidney tissue (i.e., "next-generation biopsy assessment"). They provide a novel glimpse into the pathologic processes underlying a patient's kidney disease progression, opening the possibility of tissue-based precision kidney medicine. While promising, the clinical utility (and costs) of these methods needs to be elucidated in dedicated studies.

CONCLUSION

Percutaneous renal biopsy is generally safe if care is taken to select and prepare patients beforehand. It has become a cornerstone of nephrologic practice, and its handling and interpretation should be made by those experienced in renal pathology. The interpretation of the biopsy should be carried out with adequate clinical information for integrated clinicopathologic conclusions to be drawn.

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QUESTIONS

1. Relative contraindications to percutaneous renal biopsy include the following:
   a. Uncontrolled hypertension
   b. Bleeding diathesis
   c. Hydronephrosis
   d. Single kidney
   e. Uncooperative patient

Answer: d

Rationale: All the others are absolute contraindications (see Table 26.1).

2. The best stain for viewing glomerular architecture using light microscopy is:
   a. Hematoxylin-eosin (H&E)
   b. Periodic acid–Schiff (PAS)
   c. Silver methenamine
   d. Masson trichrome
   e. Congo red

Answer: b

Rationale: PAS distinguishes the mesangial matrix and glomerular basement membrane from the other components of the glomerulus and therefore allows a good overview of the overall structure and the assessment of most lesions.

3. Pathology that affects only part of a glomerulus is termed:
   a. Segmental
   b. Mild
   c. Limited
   d. Focal
   e. Sclerosis

Answer: a

Rationale: A lesion that affects only part of a glomerulus (i.e. with some capillary lumens remaining uninvolved) is called "segmental."

4. Electron microscopy is not essential for making a diagnosis of:
   a. Alport syndrome
   b. Thin membrane lesion
   c. Immunoglobulin A nephropathy
   d. Immunotactoid glomerulopathy
   e. Fabry disease

Answer: c

Rationale: All other diagnosis rely on EM; see Box 25.5.

5. Which pathologic lesion is not found in glomeruli?
   a. Sclerosis
   b. Casts
   c. Extracapillary proliferates
   d. Amyloid
   e. Thrombi

Answer: b

Rationale: Casts are only found in tubules. All other pathologic lesions can be (also) found in glomeruli.